MATHEMATICS (SCIENCE)
Tenth, 2024 — Lahore Board — Group I
Time: 2.10 hrs. Essay Type Marks: 60

PART – I

Q.2 Write short answers to any SIX (6) questions:
12
(i) What is meant by radical equation?
(ii) Write the quadratic equation in standard form:
$x/(x+1) + (x+1)/x = 6$
(iii) Solve by factorization: $x^2 - x - 20 = 0$
(iv) Find the discriminant of the given quadratic equation:
$16x^2 - 24x + 9 = 0$
(v) Evaluate: $\omega^{-13} + \omega^{-17}$
(vi) Write the quadratic equation having following roots: 1, 5
(vii) If $R \propto T^2$ and $R = 8$ when $T = 3$, find k
(viii) Define ratio and give one example.
(ix) Find a third proportion to: 6, 12

Q.3 Write short answers to any SIX (6) questions:
12
(i) What is meant by an identity?
(ii) Convert into proper fraction: $6x^4 / (x^3 + 1)$
(iii) If A and B are two sets, then represent A – B in set builder notation.
(iv) If $Y = Z^+$, $T = O^+$, then find $Y \cup T$
(v) Find number of elements in $Y \times X$, if $X = \{a, b, c\}$ and $Y = \{d, e\}$
(vi) Find $A \times B$ if $A = \{a, b\}$, $B = \{c, d\}$
(vii) Define median.
(viii) Using logarithmic formula, find the geometric mean of 2, 4, 8
(ix) Find range: 11500, 12400, 15000, 14500, 14800

Q.4 Write short answers to any SIX (6) questions:
12
(i) Define coterminal angles.
(ii) Convert $-225°$ to radian.
(iii) Find r, when $\theta = 75°$, $\ell = 52$cm.
(iv) Show that: $\sec\theta - \cos\theta = \tan\theta \sin\theta$
(v) Define exterior of a circle.
(vi) Define circumference.
(vii) Define cyclic quadrilateral.
(viii) Draw a circle of radius 4 cm passing through points A and B, 5 cm apart.
(ix) Define escribed circle.

PART – II
Note: Attempt any THREE questions in all. But question No. 9 is Compulsory.

Q.5 (a) Solve the equation $x^2 - 3x - 4 = 0$ by completing square.
(b) Prove that: $x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x + \omega y + \omega^2 z)(x + \omega^2 y + \omega z)$

Q.6 (a) Using theorem of componendo-dividendo, find the value of
$(m + 5n)/(m - 5n) + (m + 5p)/(m - 5p)$ if $m = 10np/(n + p)$
(b) Resolve into partial fractions: $(3x + 7) / [(x + 3)(x + 4)]$

Q.7 (a) If $U = \{1, 2, 3, 4, ..., 10\}$, $A = \{2, 4, 6, 8, 10\}$, $B = \{2, 3, 5, 7\}$, then verify $(A \cup B)' = A' \cap B'$
(b) Find mean:

Classes:	33-40	41-50	51-60	61-70	71-75
No. of students:	28	31	12	9	5

Q.8 (a) If $\cos\theta = -2/3$ and terminal arm of the angle θ is in quadrant III, find the values of remaining trigonometric functions.

(b) Inscribe a circle in a triangle ABC with sides:
$|AB| = 6cm$, $|BC| = 3cm$, $|CA| = 4cm$

Q.9 Prove that a straight line drawn from the centre of a circle to bisect a chord (which is not a diameter) is perpendicular to the chord.

OR

Prove that the measure of a central angle of a minor arc of a circle is double that of the angle subtended by the corresponding major arc.